6 The polynomial $x^3 - 4x^2 + 10x - 21$ is denoted by $\mathrm{f}(x)$.

(a) Use the factor theorem to show that $(x-3)$ is a factor of $\mathrm{f}(x)$. **[2]**

(b) The polynomial $\mathrm{f}(x)$ can be written as $(x-3)(x^2+bx+c)$ where b and c are constants.

Find the values of b and c. **[2]**

(c) Show that $x = 3$ is the only real root of the equation $\mathrm{f}(x) = 0$. **[2]**